

Limits of Accuracy

Worksheet 1 (Non-Calculator)

1. w15/qp12/q9

At an athletics event, Dave and Ed each threw a javelin. Dave threw 60m, correct to the nearest 10 meters. Ed threw 61m, correct to the nearest meter.

- a. Write down the lower bound for the distance thrown by Dave. [1]
- b. Calculate the greatest possible difference between the distance thrown by Dave and the distance thrown by Ed. [1]

2. W16/qp12/q19

A box has a mass of 1.7kg, correct to the nearest 0.1kg.

- a. Write down the lower bound for the mass of the box. [1]
- b. The box holds 100 jars. Each jar has a mass of 140 grams, correct to the nearest 10 grams. Calculate the lower bound of the total mass of the box and 100 jars. Give your answer in kilograms. [2]

3. S17/qp11/q8(b)

A runner completes a race in 12.3 seconds, correct to the nearest tenth of a second. What is the lower bound for the runner's time? [1]

4. S17/qp12/q6

A thermometer measures temperature correct to the nearest degree. The outside temperature is measured as -8°C .

- a. Write down the upper bound of the outside temperature. [1]
- b. The inside temperature is measured as 10°C . Calculate the lower bound of the difference between the outside temperature and the inside temperature. [1]

5. W17/qp12/q9

The area of a rectangle is given as 8 cm^2 , correct to the nearest cm^2 .

- a. Write down the lower bound for the area of the rectangle. [1]
- b. The width of the rectangle is given as 2cm, correct to the nearest cm. Calculate the lower bound for the length of the rectangle. [1]

6. S18/qp12/q6

A rectangle has length 64mm and width 37mm each correct to the nearest millimeter.

- a. Write down the lower bound for the length. [1]
- b. Calculate the lower bound for the perimeter of the rectangle. [1]

7. W19/qp12/q6(b)

The size of a television screen is given as 95cm, correct to the nearest 5cm. Write down the upper bound of the size of the television screen. [1]

8. W20/qp11/q4(a)

Write the number 3456.789 correct to the nearest 100. [1]

9. S21/qp11/q22

A car has a mass of 2400 kg, correct to the nearest hundred kilograms. A caravan has a mass of 1460kg, correct to the nearest ten kilograms. Calculate the lower bound for the total mass of the car and caravan. [2]

10. S21/qp12/q14

- a. The length of a path is measured as 62m, correct to the nearest meter. Write down the upper bound for the length of the path. [1]
- b. The mass of a bag of peanuts is 80g, correct to the nearest 10 grams. Calculate the lower bound for the mass of 5 of these bags of peanuts. [2]

Answer Key- Worksheet 1

1. (a) 55 (b) 6.5
2. (a) 1.65 (b) 15.15
3. 12.25
4. (a) -7.5 (b) 17
5. (a) 7.5 (b) 3
6. (a) 63.5 (b) 200
7. 97.5
8. 3500
9. 3805
10. (a) 62.5 (b) 375



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